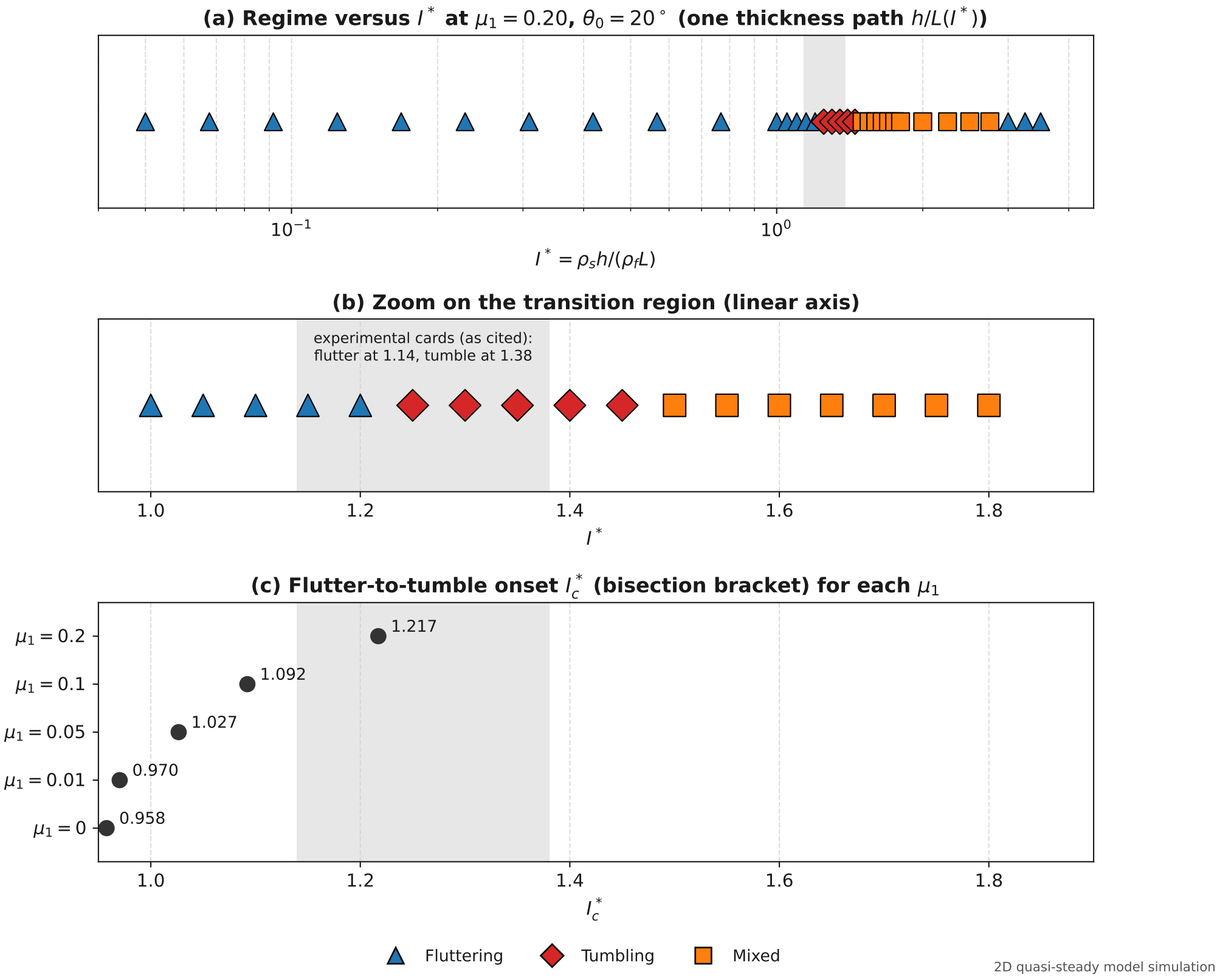
(a) Regime versus I^* at $\mu_1 = 0.20$, $\theta_0 = 20^\circ$ (one thickness path $h/L(I^*)$)
10^{-1}
10^{0}
$I^* = \rho_s h/(\rho_f L)$
(b) Zoom on the transition region (linear axis)
experimental cards (as cited):
flutter at 1.14, tumble at 1.38
1.0
1.2
1.4
1.6
1.8
I^*
(c) Flutter-to-tumble onset I_c^* (bisection bracket) for each μ_1
$\mu_1 = 0.2$
$\mu_1 = 0.1$
$\mu_1 = 0.05$
$\mu_1 = 0.01$
$\mu_1 = 0$
1.217
1.092
1.027
0.970
0.958
1.0
1.2
1.4
1.6
1.8
I_c^*
Fluttering
Tumbling
Mixed
2D quasi-steady model simulation